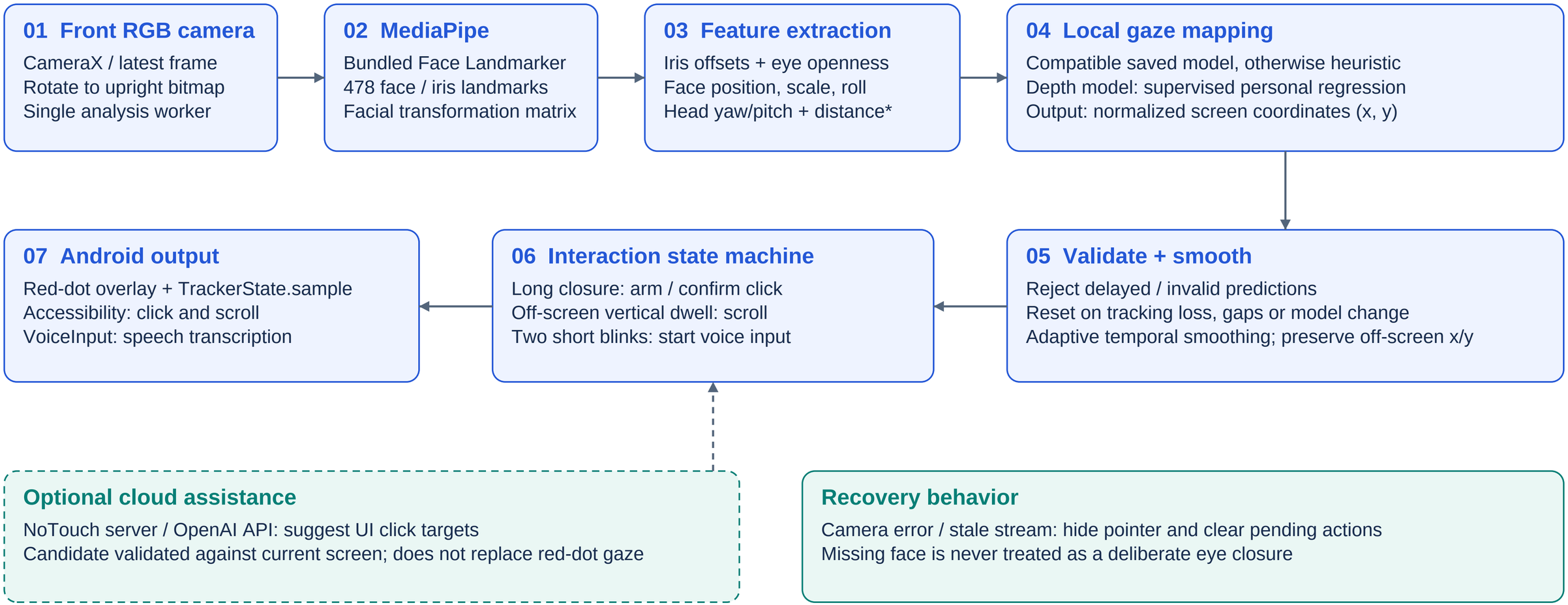
NoTouch | Runtime pipeline

ITERATION 2 / P4 Kim Gyulim + Anna | Implemented RGB path; physical-device validation pending

LOCAL ANDROID PATH / no network required for the core gaze pointer



* Distance is an iris-size estimate using camera metadata and pose gating; it is not a depth-sensor measurement.

Evidence: 41 unit tests passed; debug APK built. Device accuracy, latency and interaction success remain unmeasured.

NoTouch | Personal learning & next steps

ITERATION 2 / P4 Training code implemented; user data and measured accuracy are not included

PERSONAL TRAINING / implementation exists; no pre-trained personal weights are bundled

01 Touch-labelled sessions

Look at target, then touch it
Collect stable pre-touch features
60 targets per completed session
Stored locally as numeric CSV

02 Session-level split

At least 3 complete sessions
Earlier sessions: training
Latest session: held-out testing
Same device / display setup

03 Fit two regressors

Standardized ridge regression
With / without distance feature
Head-pose features in both
Eye x distance terms in full model

04 Evaluate & save

Held-out mean / median / P95 error
Report prediction coverage
Save full distance-aware model
Improvement is not assumed

Next acceptance gate: a connected phone

Measure horizontal response, fixation jitter, head-motion drift and distance effects
Verify click / scroll / voice actions and recovery on the actual device

Runtime model loading

ModelStore checks device / display compatibility
Local model feeds runtime gaze mapping

DEFERRED HARDWARE PATH / design only; not implemented or validated

Future: USB IR camera

Camera not acquired
Android capture / eye quality unverified

Future: synchronized RGB + IR

IR pupil / iris features + RGB head pose
Clock alignment and mount calibration needed

Future: retrain and compare

New feature schema and paired training data
Do not reuse RGB weights for IR pixel features

Touch location is a proxy label for gaze, not independently measured ground truth. Use held-out sessions for evaluation.

Current artifacts: CaptureActivity, DepthTraining, DepthModel, ModelStore. No IR acquisition, iris-texture tracker or fusion code.